

Credit Card Default Prediction

Bekzod Kadirov s333564, Temurbek Karimov s333565
Politecnico di Torino

Abstract—This work addresses the prediction of credit card client default as an imbalanced binary classification task evaluated with macro F1. We construct a single, leakage-safe preprocessing and modelling pipeline, engineer summary features from the repayment-status history, and compare four model families (logistic regression, random forest, histogram gradient boosting, and LightGBM) together with a soft-voting ensemble, all under identical stratified cross-validation and decision-threshold tuning. The exploratory analysis identifies the imbalance ($\approx 22\%$ default) and a strongly non-linear relationship between the most recent repayment status and default. After tuning, all models converge to a macro F1 of roughly 0.71, indicating that the limiting factor is the information content of the features rather than the choice of algorithm. A tuned LightGBM model is selected as the final submission, with a cross-validated macro F1 of 0.711, a public leaderboard macro F1 of 0.715, and an ROC-AUC of 0.7857.

I. PROBLEM OVERVIEW

The goal is to predict whether a credit card client will default on the next month's payment. Each client is described by demographic attributes (SEX, EDUCATION, MARRIAGE, AGE), the assigned credit limit (LIMIT_BAL), six monthly repayment-status codes (PAY_0, PAY_2–PAY_6), six monthly bill-statement amounts (BILL_AMT1–6), and six monthly previous-payment amounts (PAY_AMT1–6). The task is a binary classification problem (0: no default, 1: default). The development set contains 24,000 labelled clients and 23 input features; a separate evaluation set is unlabelled and scored externally.

Because the evaluation metric is *macro* F1 and the classes are imbalanced, accuracy is an inadequate target: a trivial majority-class predictor would attain roughly 78% accuracy while scoring poorly on macro F1. Throughout this work the focus is therefore on per-class behaviour, and in particular on the minority default class, which proves to be the harder one to predict.

A. Exploratory Data Analysis

Target imbalance. About 22% of clients default (Fig. 1). This is the central reason for evaluating with macro F1 and for using balanced class weights and decision-threshold tuning during modelling.

Repayment status. The most recent repayment-status field (PAY_0) is by far the strongest single predictor. The default rate remains low for the non-late states, then rises sharply once the client is one or more months late (Fig. 2). This non-linear threshold effect directly motivates both the encoding choices and the inclusion of non-linear models.

Numeric and demographic features. Among the numeric attributes, the six bill-statement amounts are strongly correlated with one another, while their individual correlation with the

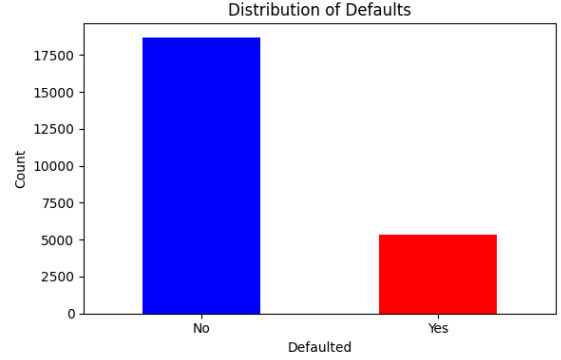


Fig. 1. Class distribution of the target. The non-default class ($\approx 78\%$) dominates the default class ($\approx 22\%$).

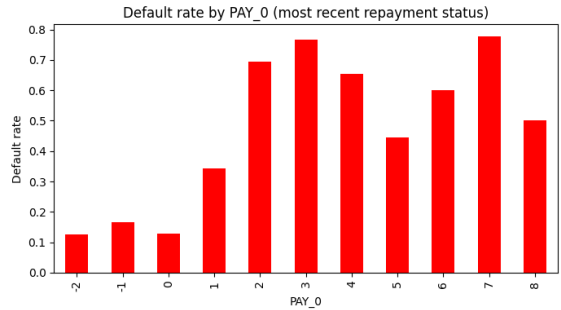


Fig. 2. Default rate as a function of the most recent repayment status PAY_0. The rate jumps once the client is one or more months late.

target is weak. Defaulting clients also tend to have lower credit limits, and demographic attributes show weaker but observable differences: men default slightly more often than women, and default rates vary moderately across education and marital-status groups. These patterns suggest that repayment history is the dominant source of predictive information, while monetary and demographic variables provide secondary signals.

II. PROPOSED APPROACH

A. Preprocessing

All preprocessing is implemented as one data-loading function together with a `scikit-learn` `ColumnTransformer` [1], so that exactly the same transformations are applied to the development and evaluation sets, and so that every fitted step is learned inside cross-validation only. Wrapping the transformer and the classifier in a single pipeline guarantees that no information from a validation fold leaks into training.

Categorical cleaning. The EDUCATION field contains undocumented codes (0, 5, 6) in addition to the four documented categories; these are folded into the “other” category (4). The undocumented MARRIAGE code 0 is folded into “other” (3). Both transformations are stateless and safe to apply before splitting.

Repayment-status fields. The six PAY_* columns contain the documented delay codes (−1 to 9) as well as the undocumented values −2 and 0. Since these values are frequent and the columns are inherently ordinal, they are retained and treated as ordinal numeric features for tree-based models.

Numeric and categorical encoding. Numeric features are median-imputed and standardised. The demographic categorical SEX, EDUCATION, and MARRIAGE are one-hot encoded, with unknown categories handled gracefully so that categories absent from a training fold do not cause failures at prediction time.

Feature engineering. Motivated by the EDA finding that repayment status carries most of the signal but is dispersed across six columns, the history is condensed into four row-wise summary features:

- **num_late**: number of months in which the client was at least one month late;
- **max_delay**: the worst delay across the six months;
- **mean_delay**: the average delay level;
- **delay_trend**: PAY_0 − PAY_6, positive when repayment is worsening over time.

These features are derived only from existing columns and use no data-derived statistics, so they remain leakage-safe and are computed identically on the development and evaluation sets.

B. Model Selection

Four model families were evaluated, chosen to span complementary inductive biases: logistic regression, random forest, histogram gradient boosting, and LightGBM [2]. All classifiers use balanced class weights to counter the imbalance. A soft-voting ensemble of HGB and logistic regression was also evaluated, on the rationale that averaging probabilities from models with different error structures can reduce variance.

The encoding of the PAY columns interacts with the model family. For tree-based models, ordinal-numeric encoding is appropriate because trees can split directly on the natural ordering. A linear model, in contrast, cannot represent the threshold effect of Fig. 2 from a single scaled column. Therefore, logistic regression was also evaluated with the PAY columns one-hot encoded, allowing the model to learn a separate coefficient for each repayment-status level. As shown in Table I, this encoding substantially improved the logistic-regression macro F1.

C. Hyperparameter Tuning

All models were evaluated with stratified 5-fold cross-validation scored on macro F1, so that each fold preserves the class ratio. The tree-based models were tuned with randomised search over their principal hyperparameters, while logistic

TABLE I
EFFECT OF PAY ENCODING ON LOGISTIC REGRESSION.

PAY encoding	Tuned macro F1
Ordinal numeric, scaled	0.630
One-hot encoded	0.706

TABLE II
EFFECT OF FEATURE ENGINEERING ON TUNED MACRO F1.

Model	Before FE	After FE
Logistic regression	0.630	0.677
Random forest	0.709	0.708
Hist. gradient boosting	0.709	0.707
LightGBM	0.695	0.711
Ensemble	0.709	0.710

regression was tuned over the inverse regularisation strength C .

For LightGBM, the search space included the number of estimators, learning rate, number of leaves, minimum child samples, L_2 regularisation, subsampling, and feature subsampling. The selected configuration used a learning rate of 0.01, 400 estimators, 63 leaves, 20 minimum child samples, and both row and column subsampling set to 0.8.

Because macro F1 depends on the decision threshold, and the default 0.5 cut-off is rarely optimal under imbalance, a final threshold-tuning step was applied. Out-of-fold predicted probabilities were swept over a grid of thresholds, and the threshold maximising macro F1 was retained. In addition to macro F1, ROC-AUC is reported as a threshold-independent measure of ranking quality.

III. RESULTS

Table II reports the effect of the engineered features on the cross-validated, threshold-tuned macro F1 for each approach. The effect is small and mixed: the engineered summaries clearly help the linear baseline and improve LightGBM, but leave the other tree models essentially unchanged. This is consistent with the fact that tree ensembles can already approximate such summaries internally through repeated splits, whereas a linear model cannot.

Table III compares the final models, after feature engineering, on cross-validated macro F1, threshold-tuned macro F1, and ROC-AUC. All models cluster tightly around a macro F1 of 0.71 and an ROC-AUC of 0.78, with LightGBM marginally ahead on every metric.

The selected LightGBM model uses a decision threshold of 0.63. It obtained a public leaderboard macro F1 of 0.715, which is close to the local out-of-fold macro F1 of 0.711 and suggests that the validation protocol generalised well to the external split.

Table IV reports the selected model’s per-class behaviour. The non-default class is predicted well ($F1 = 0.88$), whereas the default class remains harder to identify ($F1 = 0.54$). Because macro F1 averages the two classes equally, the minority class is what limits the overall score.

TABLE III

FINAL MODEL COMPARISON AFTER FEATURE ENGINEERING. CV: CROSS-VALIDATED MACRO F1; TUNED: MACRO F1 AT THE TUNED THRESHOLD.

Model	CV	Tuned	ROC-AUC
Random forest	0.707	0.708	0.781
Hist. gradient boosting	0.693	0.707	0.783
Ensemble	—	0.710	0.784
LightGBM	0.696	0.711	0.7857

TABLE IV

PER-CLASS METRICS OF THE SELECTED LIGHTGBM MODEL.

Class	Precision	Recall	F1	Support
0	0.86	0.90	0.88	18691
1	0.58	0.50	0.54	5309
Macro avg	0.72	0.70	0.71	24000

IV. DISCUSSION

The most informative result is not the absolute score but the convergence: four model families spanning linear, bagged, and boosted approaches all reach a macro F1 of about 0.71 after tuning, and their ROC-AUC values are likewise grouped around 0.78. When algorithms with such different inductive biases agree this closely, they are unlikely to be limited by their individual capacities; rather, they are bounded by the information available in the features.

Three observations support this interpretation. First, hyperparameter tuning moved the boosted models only marginally, indicating that their defaults were already close to optimal for this representation. Second, the engineered repayment summaries produced only a small lift, mainly for the linear model; the tree models were essentially unchanged because they can already form equivalent splits internally. Third, the limiting factor across all models is the default class, whose F1 remains at 0.54.

The threshold-tuning step gave a consistent, if modest, improvement over the default cut-off and is well justified for a macro-F1 objective under imbalance. The soft-voting ensemble did not meaningfully exceed the best single model, which is expected when its members are individually strong but highly correlated. For these reasons the final submission uses the single tuned LightGBM model: it is marginally the strongest on every metric while remaining simpler and faster than the ensemble.

Limitations and future work. The analysis is bounded by the available features. The recorded six-month window does not include external events that may precipitate default, which places a ceiling on achievable separation of the minority class. Within this constraint, the most promising direction is additional feature representation rather than further model search, for example credit-utilisation ratios (BILL_AMT/LIMIT_BAL), payment-coverage ratios (PAY_AMT/BILL_AMT), and additional trend features over the six months.

REFERENCES

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- [2] G. Ke *et al.*, “LightGBM: A Highly Efficient Gradient Boosting Decision Tree,” in *Advances in Neural Information Processing Systems*, 2017.